Tactix certified GF(2) organic integration
Current evidence and corrected probe

Development report for Andrew
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Disposition

The preservation baseline is healthy. Immutable source d7e48970d passes the complete Debug campaign and all four standing regressions, with no wrong verdict, checker rejection, or scorer failure. Subsequent opt-in GF(2) work supplies a learner-selectable certified Boolean action, but its first target experiment recovered no target: all three completed rows timed out before an affine action was offered. This report makes no fixed-target, performance, or portfolio-recovery claim.

Campaign	Disposition	Solver time	Independent disposition
Debug	292/292	820.39 s	244 executables; 340 immutable bindings
B28+C5	33/33	959.350 s	32 checked; one clean proof censor
Rank1--50	50/50	7,620.029 s	42 checked; eight clean proof censors
WV100	100/100	2,806.871 s	98 checked; two clean proof censors
WV2--89	89/89	4,029.436 s	78 checked; two clean censors; nine allowed timeouts

B28+C5 is below its 999.87615-second ceiling and Rank is below 8,521.52175 seconds. Every proof censor is a clean expiration of the 1,800-second checker allowance. The accepted jobs are Debug 1750129, B28+C5 1750135, Rank 1750168, WV100 1750241, and WV2 1750302. They are the current preservation authority. The controller and telemetry code changed afterward, so the corrected checkpoint still owes fresh Debug and four-gate qualification.

What changed after the accepted checkpoint

- a7cd75ba5 adds a supplemental fail-closed GF(2) target auditor. Missing or capped telemetry cannot be scored as success.

- 1ea10b1b5 makes a completed empty/no-structure census terminal for its immutable owner across search and restart epochs. Explicit resource/interruption retry is preserved. Semantic advice is deferred until the configured propagation follow-up finishes; fixed/retired variables are filtered and there is no hidden fallback.

- 13eea1458 makes the named search-configuration subset learner-owned and replay-fingerprinted. A probe can hold one configuration while exposing AffineConsequence, explicit Propagate, and nonzero SemanticVariableAdvice.

- fa0f0d5b6 records the initial target diagnosis and corrected contract in the progress, calculus, interface, and implementation documents.

Focused tests cover durable no-structure suppression, bounded paging, explicit resource retry, post-propagation VMTF/EVSIDS advice, stale rejection, deterministic replay, and fail-closed configuration. This establishes mechanism behavior, not target recovery.

Initial target evidence

Jobs 1750437 and 1750438 used d7e48970d, DSR, the gf2-affine producer, and the original one-action profile. Their anonymous runtime validations passed.

Row	Limit	Calls	Work	Candidates/offers
WV2-132 tseitin_d3_n158 UNSAT	600 s	397	4,789,262	0/0
WV100-7 two-trees-511v SAT	1,200 s	281	1,940,937	0/0
WV100-51 mdp-32-11-sat SAT	1,200 s	8	170,560	0/0

All three returned UNKNOWN at the solver limit. No checker ran and no X9 request was made. The retained statistics establish calls and zero offers, but not a complete action chain. The old format wrote a full cumulative observation around millions of search-service actions, exceeded the 64 MiB cap, and deleted the feedback file. The supplemental auditor correctly rejects every row. Compact, lossless, versioned feedback is under redesign; observation must stay pure and preserve the raw meanings needed by a future learner.

Diagnosis and corrected probe

The causes are concrete. WV2-132 reaches image width 68 over 157 sources and has 130 selectable rows, beyond old 32/32 bounds. WV100-7 reaches width 62 over 160 sources and has 326 selectable rows. WV100-51 has 2,452 groups against the old 1,000-group cap and exposes 500 selectable rows after that cap is raised. The old controller also sampled all 20 named search configurations. Durable census and the search allowlist repair the two control defects.

Fixed-rank job 1750439 is still running on ranks 263 and 569 under the old profile. It contributes no completed evidence.

Corrected four-row probe

The next immutable probe contains exactly WV2-132, WV100-7, WV100-51, and fixed rank 263. It allows one named search configuration and AffineConsequence, Propagate, and SemanticVariableAdvice. The bounded GF(2) profile is:

```
affine-consequence2(12000,30000,1000000,2500,10000,10000000,8,
68,160,4,1000000,1000000,10000000,10000000,68,160,10000,10000,
3,100000,10000,1000000)
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This raises maximum_groups to 2,500 and image/basis width/source limits to 68/160 while retaining finite proof, journal, extent, pattern, and work bounds. Rank 263 has 1,899 groups and 1,328 selectable rows.

Exclude rank 569. Its census has 10,158 groups and 10,230 patterns, but only 24 independent one-source width-three equations and no selectable consequence. Raising the profile would add census cost without exercising X9. The running old job does not change this unless it produces contradictory evidence.

Every corrected row must retain complete telemetry: producer status/effort, candidate, offer, request and application counts, proof work, extent, yield, and terminal state. Credit requires an independently checked model or proof. Timeout, checker failure, missing sidecar, cap loss, or inconsistent terminal records remain fail-closed diagnostic outcomes.

Soundness, scheduling, and learning readiness

The integration boundary remains organic: GF(2) reasoning authenticates theory structure and offers a typed Boolean consequence or conservative extension to continuing CDCL. It is neither a private local solver nor a sequential portfolio. DSR remains the default proof format.

For N1, the exact affine row is an ordinary public action; search configuration, propagation, and advice are explicit shared-controller choices. For N2, target credit requires an independently checked reconstructed model or proof, with extension projection and provenance. For N3, the accepted default is competitive, but post-fix source must repeat all four gates. For N4, order, effort, retry status, search configuration, and advice are typed and observable; the scheduling ledger and final policy acceptance remain pending.

Default, random, and future learned policies share action identities and the execution path. Pure observations and after-action raw feedback expose effort and result without rollback or hidden fallback. Next: finish lossless telemetry; freeze and validate the corrected four rows; require checked outcomes; then run effective both-policy fuzz, fresh Debug, and all four regressions. Expand to ranks 540/896/1004 and MAB-only 557 only after coherent checked evidence. The saved d170b4a5 comparison remains authoritative and is not rerun.